WATERWATCH · PROJECT PLAN

WaterWatch project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	O1 to O5 decided by Amish (2026-10-25); calibration kit with the shared photometer
0.1	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: workstreams, resources, stage gates, decision request and the reference list

WaterWatch project plan

WaterWatch is at stage P3 (design outputs, TRL 3) with every P0 to P3 deliverable in place; the next gate, P4 verification, is on hold under the portfolio's stage cap of P3. Amish decided the concept carried forward (option A), the calibration reference graft, the product render layout, how water use counts event flushes and the chlorine accuracy target above 1 mg/L on 2026-10-25 (WWT-DEC-001 O1 to O5). Still open: the approach to the first candidate partner, which is Amish's outreach. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Project lead	WWT-PRB-001 with users and stakeholder tables	Done (P0 to P1)
STEM check	Design lead	WWT-STM-001 and `docs/04-calcs/stem_check.py`	Open: 9 plausible rows for option A need tests (TRL 4); S31 awaits Amish (O5)
Concept options	Design lead	WWT-OPT-001, WWT-DDR-004, D3 models of A, B and C	Done; option A chosen by Amish, 2026-10-25
Requirements and traceability	Design lead	WWT-URS-001, WWT-PRS-001, WWT-RTM-001	Done (P2)
Calculations	Design lead	WWT-CAL-001 v0.5	Done (P2 to P3)
Design outputs	Design lead	`cad/src/model.py` at D3, STEP, drawings WWT-DWG-001 and 101 to 109, priced BOM, WWT-BLD-001, WWT-DEC-001	Done (P3)
Regulatory, feasibility, risk, market	Project lead	WWT-REG-001, WWT-FEA-001, WWT-RSK-001 with design FMEA, WWT-MKT-001	Done (P3 first pass)
Partner and pilot	Project lead	Approach the first candidate partner (Water Mission, WWT-DEC-001)	Not started; Amish's decision to approach
Verification (P4)	Design lead with a lab	Bench drift test of the chlorine sensor, turbidity standards, shield temperature log, backflow test	On hold (stage cap P3)
Validation (P5)	Partner	Field trial at two or three tapstands	On hold (stage cap P3)

Resources needed

Table 2. Resources for the next stages (estimates, for planning only; no spending is proposed while the cap is P3).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts	One unit to the BOM	Workshop build, about 3 days	325 (BOM, indicative)
Calibration kit	Handheld DPD photometer (O2, decided 2026-10-25), pH comparator, turbidity standards	Shared per operator	380 to 600 for the photometer (WWT-MKT-001), plus about 60 for the comparator kit (estimate)
Lab bench time	Chlorine drift test against DPD over 4 weeks or more (closes S12, S14)	About 2 days set-up, 4 weeks logging	To assess with a partner lab
Partner time	Site survey (pressure, pH, coverage), caretaker routine	2 to 3 site visits	To assess with the partner
Regulatory advice	Radio type approval route, wetted materials (WWT-REG-001 R7, W1)	Consultant, a few days	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met
P1 Definition	URS, PRS, RSK, MKT	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR	Met; the DDR's choice awaits Amish
P3 Design outputs	Model at D3 for every option, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability complete	Met (`cad/fidelity.json` passed)
P4 Verification	Bench tests close S4, S5, S12, S14, S22, S32 to S34, S36; PR-001, PR-002, PR-009, PR-010 verified	On hold (stage cap P3)
P5 Validation	Field trial with a partner; certification status in REG	On hold
P6 Commercialization	Launch plan with a named deployment partner	On hold

Next steps and decision requested

1. Done: Amish decided O1 to O5 on 2026-10-25 as recommended (WWT-DEC-001): option A, the shared photometer, installed heights in the product render, PR-008 counting the hourly schedule only, and PR-001 restated as ±30 % above 1.0 mg/L.
2. With his go, approach the first candidate partner with this TRL 3 pack (WWT-DEC-001, 2026-10-02).
3. When the portfolio cap allows TRL 4: build one unit and run the chlorine drift test first, because it decides between A and the fallback B (WWT-RSK-001 P1, F1).

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